INORGANIC CHEMISTRY

UNIT 3 NOTES

GASTROINTESTINAL AGENTS

- ACIDIFIERS
- ANTACIDS
- CATHARTICS
- ANTIMICROBIALS

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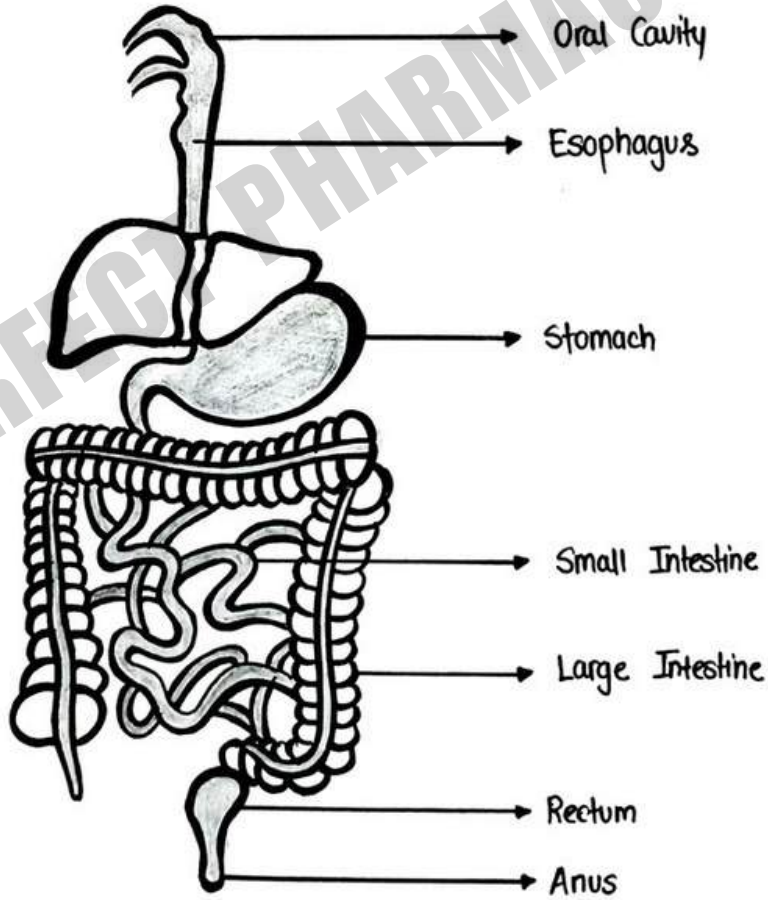
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GASTROINTESTINAL TRACT

- The gastrointestinal tract is the pathway of digestive system which helps in the digestion of food and its excretion.
- It starts from the mouth and end to the anus.
- The main portion of GIT includes the stomach, small intestine, large intestine, rectum and anus.
- Now, whenever the function of GIT goes wrong, disease occurs.



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Diseases Of GIT

- Insufficient secretion of acid in the stomach can cause Achlorhydria or Hypochlorhydria.
- Too much secretion of acid in the stomach can cause Hyperacidity.
- The insufficient bowel movement in the intestine can leads to Constipation
- Also the growth of microbes / bacterias can affect the Gastrointestinal Tract.

GASTROINTESTINAL AGENTS

- Agents that are used to treat gastrointestinal disturbance / diseases are known as Gastrointestinal Agents.
- Agents that are used to GIT disorders includes :

① Acidifiers (Hypochlorhydria)
② Antacids (Hyperacidity)
③ Cathartics (Constipation)
④ Antimicrobials (Microbial Growth)

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ACIDIFIERS

- These are the inorganic substances that are used to increase the level of gastric acid in the stomach.
- They are also known as Acidifying Agents.
- They either produce acid or increase the level of acid.
- They work by decreasing the pH of stomach.
- They are used in the treatment of Achlorhydria or Hypochlorhydria.

Why Acid is Important in the Stomach ?

- It breakdown the food and help in digestion.
- Kill microbes present in the food.
- Provide acidic environment needed for effective digestion.

Types Of Acidifiers

There are mainly 3 types of Acidifiers

① Gastric Acidifiers
② Urinary Acidifiers
③ Systemic Acidifiers

Gastric Acidifiers

- They are mainly used to control the pH of stomach.
- These are the drugs which are used to increase the acidity of stomach.
- These agents are mainly used in patients suffering from Achlorhydria or Hypochlorhydria.

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Urinary Acidifiers

- They are mainly used in controlling the pH of urine.
- These are the drugs which are used to remove acidic urine from the body.
- These acidifiers are widely used to cure some type of Urinary Tract Infection.

Systemic Acidifiers

- They are used to control the pH of all parts of body.
- These are the drugs which are able to neutralized the alkaline body fluid.
- It is used to treat patient suffering from Alkalosis.

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Ammonium Chloride

Molecular Formula : NH_4Cl
Molecular Weight : 53.49 g/mol
Synonym : Sal Ammoniac

Method of Preparation

It is prepared by the reaction of HCl with NH_3 (ammonia)

$$NH_3 + HCl \longrightarrow NH_4Cl$$

Physical Properties

- It is a white crystalline powder
- It is odourless.
- It has cool saline taste
- It is hygroscopic in nature

Chemical Properties

On decomposition ammonium chloride produces ammonia gas and hydrogen chloride.

$$NH_4Cl \longrightarrow NH_3 + HCl$$

Uses of Ammonium Chloride

- It is used as Acidifiers.
- It is used as Fertilizers.
- It is used in Glue.
- It is also used in Buffer Solutions.

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DILUTE HCL

Molecular Formula : HCl
Molecular Weight : 36.46
Synonym : Spirit of Salt

Preparation

It can be prepared by action of concentrated H_2SO_4 with NaCl

$$NaCl + H_2SO_4 \longrightarrow HCl + NaHSO_4$$

Properties

- It is a colourless liquid
- It is strongly acidic
- It is soluble in water and alcohol
- It has pungent odour
- On reacting with NaOH, it gives salt and water.

$$HCl + NaOH \longrightarrow NaCl + H_2O$$

Uses

- It is used as acidifier
- It is mainly used as an pharmaceutical aid
- Used in many Solvents.

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ANTACIDS

- Antacids are the drugs that are used to decrease the level of gastric acid in the stomach.
- They are used in the case of Hyperacidity.
- They also prevent from Heart Burn and Ulcer.
- Antacids are generally weak bases that reacts with the excessive acid in the stomach and convert them into salt.
- They work by increasing gastric pH.
- example : $CaCO_3$, $NaHCO_3$

Classification of Antacids

Antacids are commonly classified into two groups :

① Systemic Antacids
② Non- systemic Antacids

Systemic Antacids

- Antacids which absorbs in the systemic circulation (blood circulation) are called as Systemic Antacids.
- They are not very suitable antacids as they can cause Metabolic Alkalosis (Disturbance in acid- base balance)

Non- Systemic Antacids

- Antacids which are not absorbed into systemic circulation and do not affect acid- base balance of the body are called as Non- Systemic Antacids

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Ideal Properties of Antacids

- It should be insoluble in water and has fine particle form.
- It should not cause metabolic alkalosis.
- It must have its effect over a long period of time.
- It should not cause constipation.
- It should not cause any side effect.
- It should be stable and readily available.
- It should not be too expensive.

Combination of Antacids

- Systemic antacids are not used regularly as they can cause metabolic alkalosis and congestive heart failure.
- Non-systemic antacids are more effective compared to systemic alkalosis, but they also have some other side effects.
- To avoid these side effect and to increase the effect and time of duration of action. antacids are generally given in combination.

- Generally in the combination of antacids, one having a rapid action and other having longer duration of action.
- Example : Combination of Magnesium and Aluminium as Antacid
  - (i) Magnesium → Fast acting
  - (ii) Aluminium → Longer duration of action

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Sodium Bi Carbonate

Chemical Formula : $NaHCO_3$
Molecular weight : 84.01 g/mol
Synonym : Baking Soda

Preparation

It is prepared by solvay ammonia process.

$$Na_2CO_3 + CO_2 + H_2O \longrightarrow 2NaHCO_3$$

Properties

- It is a white crystalline powder.
- It is insoluble in ethanol but soluble in methanol
- It having saline taste
- It is alkaline in nature
- It is also soluble in water

Uses

- It is mainly used as antacids.
- It is used as electrolyte replenisher.
- It is also used as disinfectant.

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Aluminium Hydroxide Gel

Chemical Formula : $Al(OH)_3$
Molecular Weight : 78.00 g/mol
Synonym : Aluminium Hydrated Powder

Preparation

It is prepared when Aluminium Chloride is treated with Ammonium Hydroxide

$$AlCl_3 + 3\,NH_4OH \longrightarrow Al(OH)_3 + 3\,NH_4Cl$$

Properties

- It is a white viscous suspension
- It is tasteless.
- It is odourless.
- It is soluble in mineral acid solution, but insoluble in water & alcohol.

Uses

- It is used as antacid.
- It is used to treat heartburn

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CATHARTICS

- Cathartics are the drugs that are used to get relief from constipation.
- These are the drugs that... accelerates defecation.
- Cathartics act by increasing the fluid content of Faeces, making them softer and easier to pass.
- Cathartics increases the mobility of intestine

Types of Cathartics

Cathartics are mainly classified into two categories :

① Laxative
② Purgative

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Laxative

- Laxatives are mild acting cathartics
- They work by either :
  - Increasing intestinal movement
  - Increasing stool bulk
  - Make stool softener
- Prolong use of laxative may cause habit or dependency.

Purgative

- They are strong cathartics
- They are given in very serious conditions.
- They are generally given to completely remove solid materials from intestine before surgery.

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Classification (On the basis of mechanism)

① Stimulant Cathartics
② Lubricants
③ Bulk Forming
④ Saline Cathartics

Stimulant Cathartics : They act by producing local irritation on intestinal tract.

Lubricants : Provide lubricant effect so that stool easily passes through rectum, also known as stool softener.

Bulk Forming : These agents increases the amount of stool production.

Saline Cathartics : They increases the osmotic load of GIT, consumed with large amount of water.

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Uses of Cathartics

- For easy defacation & other rectal diseases.
- To relief from acute constipation.
- To remove solid material from intestinal tract before surgery.
- To avoid rise in blood pressure due to constipation.

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Magnesium Sulphate

Chemical Formula : $MgSO_4 \cdot 7H_2O$
Molecular Weight : 246.47 g/mol
Synonym : Epsom Salt

Method of Preparation

It is obtained by the action of dilute sulphuric acid (H_2SO_4) and magnesium carbonate ($MgCO_3$)

$$MgCO_3 + H_2SO_4 \longrightarrow MgSO_4 + H_2O + CO_2$$

Properties

- It occurs as white crystals
- It is odourless
- It having a cool, saline, bitter taste
- It is soluble in water and sparingly soluble in alcohol.

Uses

- It is used as cathartics.
- It is used in agriculture.
- It is used to control seizures in pregnancy.

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Sodium Orthophosphate

Chemical Formula : $Na_2HPO_4 \cdot 12H_2O$
Molecular Weight : 358.14 g/mol

Preparation

It is obtained by adding sodium carbonate to a hot solution of phosphoric acid.

$$H_3PO_4 + Na_2CO_3 \longrightarrow Na_2HPO_4 + H_2O + CO_2$$

Properties

- It occurs as colourless crystals
- It is odourless
- It having a saline taste
- It is soluble in water
- It is insoluble in alcohol

Uses

- It is used as cathartics
- It is also used as buffers.

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Kaolin

Molecular / Chemical Formula : $Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$
Molecular Weight : 258.16 g/mol
Synonym : China Clay

Preparation

It is simply prepared from natural clay by powdering, separating and purifying process.

Properties

- It is light, white powder.
- It is odourless
- It is tasteless

Uses

- It is used as cathartics
- It is used in food poisoning
- It is used in dusting powders.

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Bentonite

Chemical Formula : $Al_2O_3 \cdot 4SiO_2 \cdot H_2O$
Molecular Weight : 360.31 g/mol
Synonym : Clay

Preparation

It occurs naturally or can be prepared from natural clay

Properties

- It occurs as very fine, cream-colour powder
- It is odourless.
- It is insoluble in water.

Uses

- It is used as cathartics.
- It is used as emulsifier.
- It is also used as protectives.

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ANTIMICROBIAL AGENTS

- Antimicrobial agents are those chemical compounds or drugs that inhibits or destroys the growth of microorganism
- They can either kill or prevents the growth of bacteria.

Classification of Antimicrobials

① Antiseptic
② Disinfectants
③ Germicides
④ Bacteriostatics
⑤ Sanitizers
⑥ Sterilization

Antiseptic

- Antiseptic are those antimicrobial agents that are mainly used on living cells.
- They either kills or inhibits the growth of bacteria.
- An ideal antiseptic agent should destroy bacteria, viruses etc and should not cause any harm to applied area.
- Example : Hydrogen Peroxide, Silver nitrate etc.

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Disinfectant

- These are the antimicrobial agents that are mainly used on Non-Living surfaces.
- They can either kill or destroy bacteria.
- They are mainly used in home and hospital cleaning.
- example : Sulpher dioxide

Geomicides

- These are the agents that are used to kill microorganism.
- The can be either use on Living or Non- living surfaces.
- They can also be divided into following types :

  Bactericides : To kill bacteria
  Virucides : To kill viruses
  Fungicides : To kill fungi

Bacteriostatics

- These are the agents that are used to prevent the growth of bacteria.
- They do not kill bacteria
- Can be used on living or Non- living surfaces.
- example : Chloramphenical

Sanitizers

- Sanitizers are agents that are mainly used in cleaning & washing.
- They are generally used to maintain general public health standards.
- example : Soap, Alcohol etc.

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Sterilization

- It is a process by which all the microorganism are either killed or removed.
- In this process, the products, surface or area will be free from all type of microorganism.

Mechanism of action of Antimicrobials

- Microorganism mainly contains proteins (enzymes) to survive.
- Anti-microbials act by changing their protein structure which results in death of microorganism.
- Antimicrobials mainly act by 3 mechanism:

① Oxidation
② Halogenation
③ Precipitation

Oxidation

- This MOA shows by oxygen releasing compounds.
- They cause oxidation of active functional groups present in proteins and inactivate them.

Halogenation

- This MOA shows by halogen (Br, Cl etc) releasing compounds.
- These agents add halide atom to nitrogen present at peptide linkage of protein & inactivates them.

Precipitation

- This MOA shows by Metal containing compounds.
- Metal binds with important group present in protein change their structure & inactivates them.

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POTASSIUM PERMANGANATE

Chemical Formula : $KMnO_4$
Molecular Weight : 158 g/mol
Synonym : Condy's Crystals

Preparation

It can be easily prepared ~~by~~ from potassium manganate under acidic condition

$$2K_2MnO_4 + 4HCl \longrightarrow 2KMnO_4 + MnO_2 + H_2O + 4KCl$$

Properties

- It is purple crystalline solid.
- It is odourless.
- It is soluble in both water and alcohol.

Uses

- It is used as antimicrobial agents.
- It is used in treatment of various skin conditions.
- It is used as antiseptic
- It is used as strong oxidizing agent.

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Boric Acid

Chemical Formula : H_3BO_3
Molecular Weight : 61.83 g/mol
Synonym : Hydrogen Borate

Preparation

It is prepared by the action of HCl on borax ($Na_2B_4O_7$)

$$Na_2B_4O_7 + 2HCl + 5H_2O \longrightarrow 2NaCl + 4H_3BO_3$$

Properties

- It is white crystalline solid
- It is odourless
- It having sweet taste.
- It is soluble in water & slightly soluble in alcohol.

Uses

- It is used as anti-microbial agents.
- It is used as preservatives.
- It is also used in cosmetics.

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HYDROGEN PEROXIDE

Chemical Formula : H_2O_2
Molecular Weight : 34.01 g/mol
Synonyms : Hydrogen Oxide , Peroxide

Preparation

It can be prepared by the reaction of sodium peroxide with dilute sulphuric acid

$$Na_2O_2 + H_2SO_4 \longrightarrow H_2O_2 + Na_2SO_4$$

Properties

- It is clear colourless liquid.
- It is odourless.
- It having a bitter taste.
- It is miscible with water.

Uses

- It is used as antimicrobial agent.
- It is used as antiseptic.
- It is used as bleaching agent.
- Also used in tooth whitening.

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CHLORINATED LIME

Chemical Formula ~~$Ca(OCl)_2$~~ $CaOCl_2$

Molecular Weight ~~142.98 g/mol~~ 136.98 g/mol

Synonym Bleaching Powder

Preparation

It is prepared by reacting chlorine with calcium hydroxide

$$Ca(OH)_2 + Cl_2 \longrightarrow CaOCl_2 + H_2O$$

Properties

- It is white or grey powder.
- It has odour of chlorine.
- It is partially soluble in water and alcohol.

Uses

- It is used as antimicrobial agent.
- It is used as powerful bleaching agent.

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IODINE

Chemical Formula : I_2
Molecular Weight : 253.8 g/mol
Synonym : Halogens

Preparation

In laboratory, it is prepared by heating the mixture of kI, MnO_2 and concentrated H_2SO_4.

$$2KI + MnO_2 + 3H_2SO_4 \longrightarrow I_2 + 2KHSO_4 + MnSO_4 + 2H_2O$$

Properties

- It occurs as greyish-violet or bluish-black crystals.
- It has strong harsh odour.
- It is volatile in nature.
- It is insoluble in water.
- It is soluble in alcohol.

Uses

- It is used as antimicrobial agent.
- It is used in iodine deficiency.
- It is used in thyroid cancer

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